The reigning paradigm in cosmology is called the Concordance Model, or ΛCDM. In this picture, the universe has four basic components: radiation, regular matter, dark matter (specifically "cold" dark matter, CDM), and dark energy in the form of a cosmological constant (denoted in equations by the Greek letter lambda, Λ). The quantities of all these components are precisely measured, with the cosmological constant currently making up the largest slice of the cosmic pie. We have a good understanding of how these things have all varied over time as the universe has expanded, and we have an amazingly detailed description of the very early universe that includes a period of very rapid expansion called inflation. We also have a tried-and-tested theory of gravity, Einstein's general relativity, which in the Concordance Model is taken to be completely correct. In this picture, because the cosmological constant is currently dominating the evolution of the cosmos, we can straightforwardly apply our understanding of gravity and the components of the universe to determine our cosmic evolution. Doing this leads us unambiguously to a Heat Death in the far future. And that's that.

The problem with the Concordance Model is that the most important elements of it—dark matter, the cosmological constant, and inflation—are completely mysterious. We don't know what dark matter is; we don't know how inflation happened (or if it even really did happen); and we have no reasonable explanation for why the cosmological constant exists or why it takes a value that seems to fly in the face of what we expect from particle physics. At the same time, we haven't found anything in the data to contradict the model. No evidence that dark energy evolves in some way (which would go against a cosmological constant), no evidence that dark matter is anything experimentally detectable (and no evidence that it's not), and despite a century of putting it through the experimental ringer, no evidence for gravity behaving like anything other than Einstein's general relativity.

Andrew Pontzen, a colleague and coauthor of Peiris's (and my former officemate at Cambridge) works on theoretical aspects of dark matter, and has done some of the pioneering work to explain why dark matter takes the shape it does in galaxies. He contends that we have a very good understanding of cosmology, in the sense that our data line up extremely well with a picture that includes dark matter and dark energy, and that it seems unlikely anything will suddenly appear to change that picture. We